

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

Never turn the engine in the direction opposite to that of normal operation

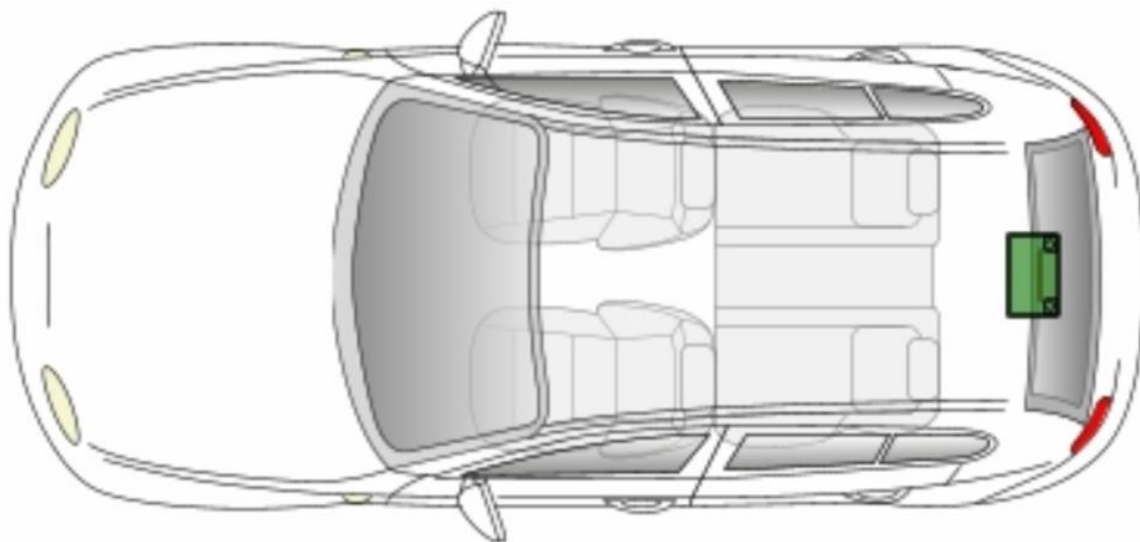
Note: Avoid bending, torsion or traction

Always use new gaskets

General

Before disconnecting the battery cable, check the audio system security code

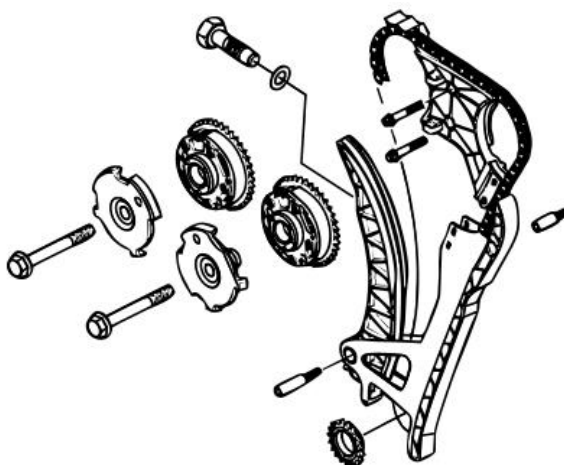
Battery location



Luggage compartment

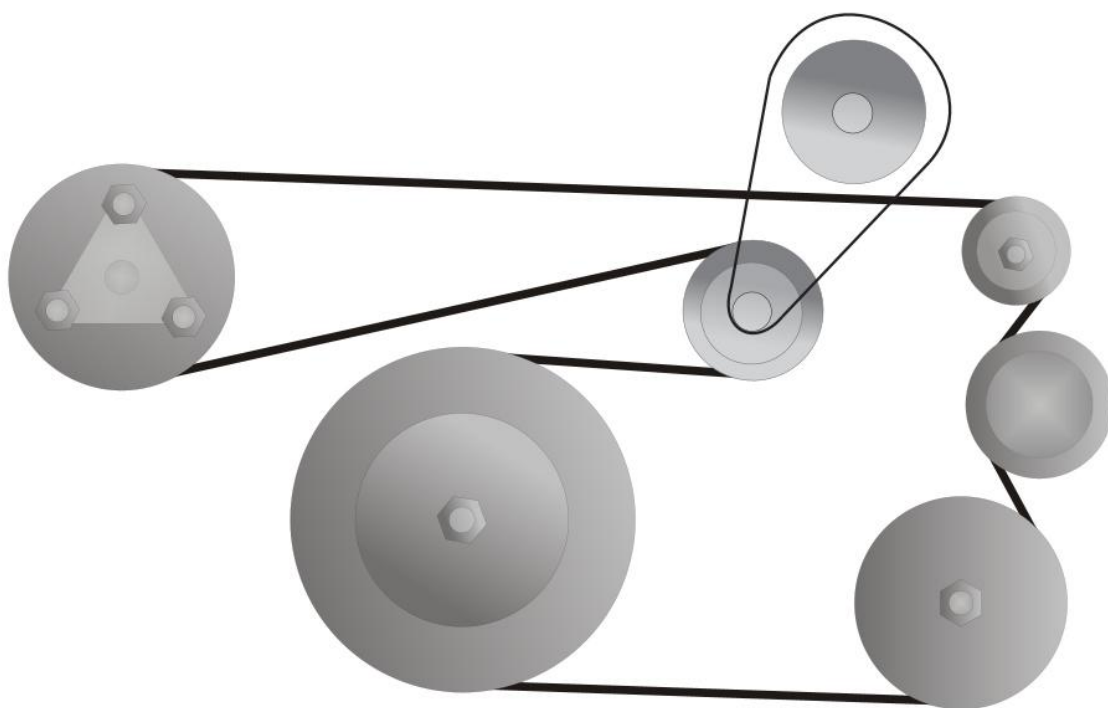
Timing chain layout

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Removal

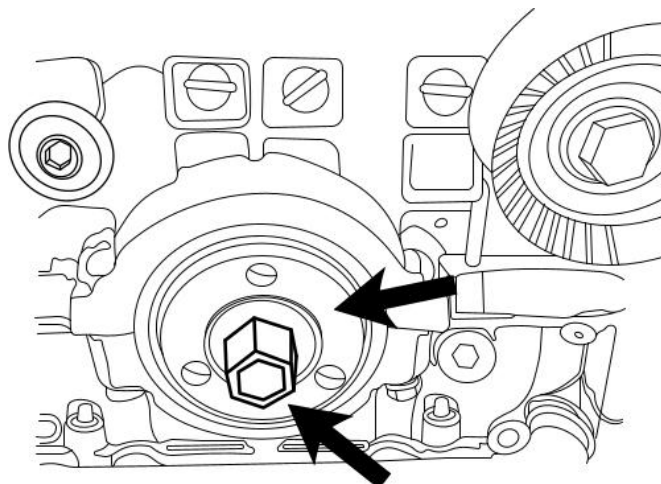
- Disconnect the battery earth cable
- Remove the engine upper cover
- Remove the ignition coils
- Disconnect the injection wiring
- Remove the strut brace connecting the shock absorbers
- Remove the air duct
- Disconnect the cylinder head breather hose
- Remove the cylinder head cover
- Remove the spark plugs
- Remove the chain tensioner piston
- Compress the chain tensioner piston twice to drain the oil
- Remove the engine lower cover
- Remove the ancillary drive belt
- Drive belt layout



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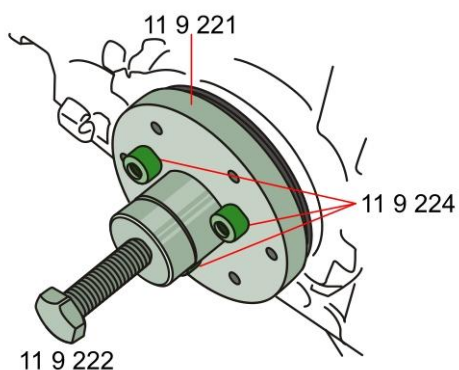
Remove the vibration damper

Note: Do not remove



Fit the special tools (11 9 221, 11 9 222, 11 9 224)

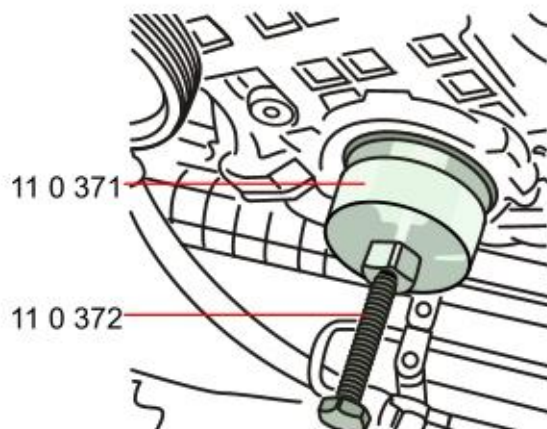
Tighten the special tool (11 9 224 - 20 Nm)



Remove the special tools (11 9 221, 11 9 222, 11 9 224)

Fit the special tools (11 0 371, 11 0 372)

Tighten the special tool (11 0 371 - 80 Nm)



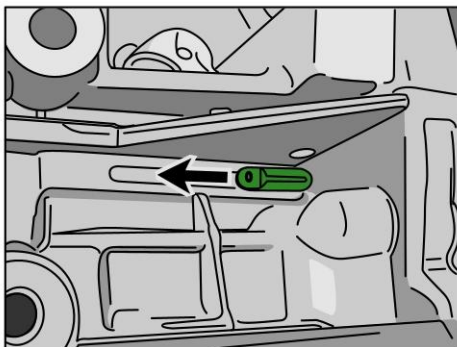
Remove and discard the crankshaft oil seal

Remove the tensioner locking pin (11 3 340)

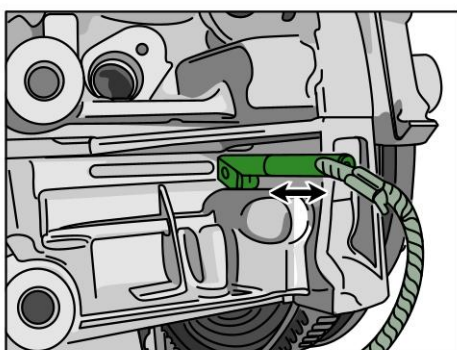
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Remove the ancillary drive belt tensioner

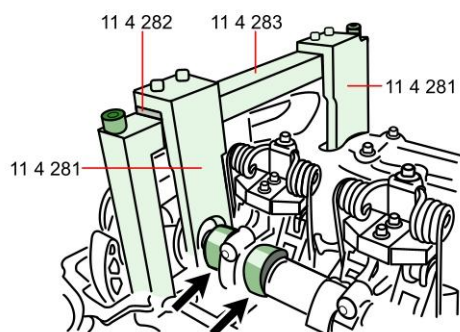
Remove the access plug



Fit the crankshaft locking tool (11 0 300)

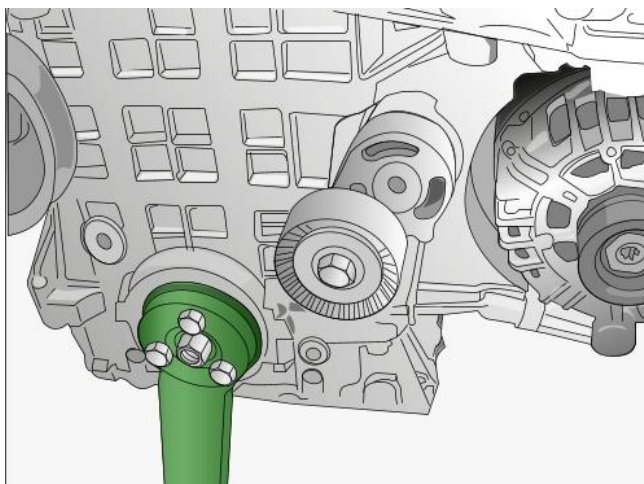


Fit the camshaft locking tools (11 4 281, 11 4 282, 11 4 283)



Remove the crankshaft gearwheel centre bolt

Use the special tool to counterhold (11 9 280)

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Remove the crankshaft hub

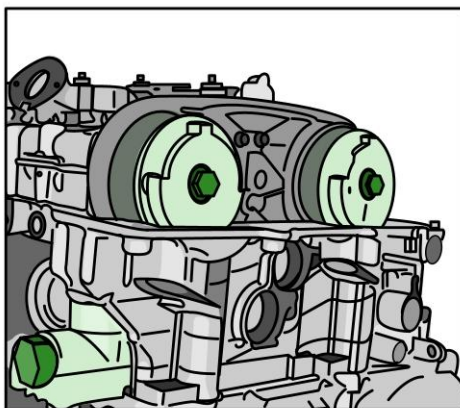
Bearing pins for guide and tensioner rails:

Remove the access plugs

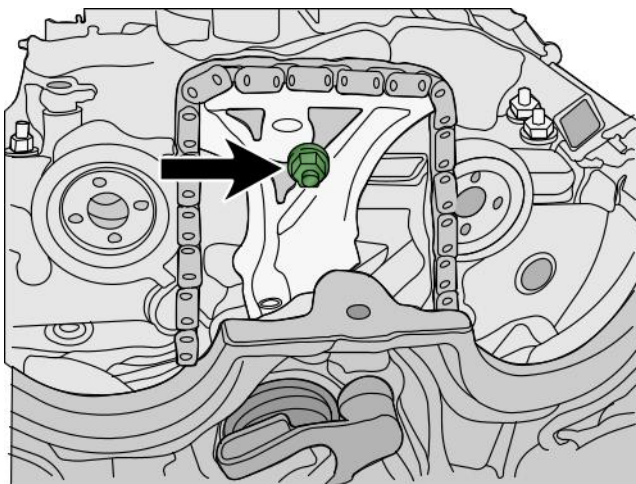
Remove the guide pins

Remove the inlet camshaft timing adjuster

Remove the exhaust camshaft timing adjuster



Remove the timing chain guide bolts from the cylinder head

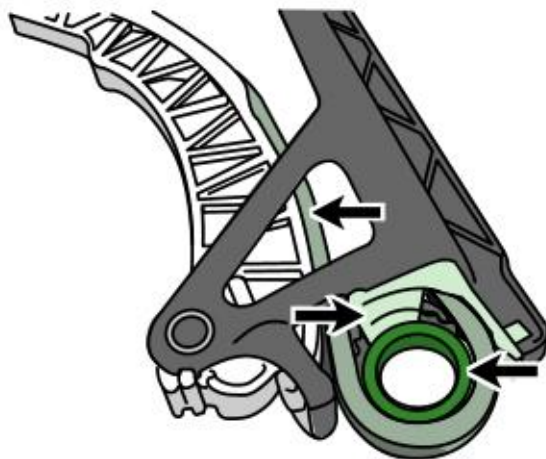


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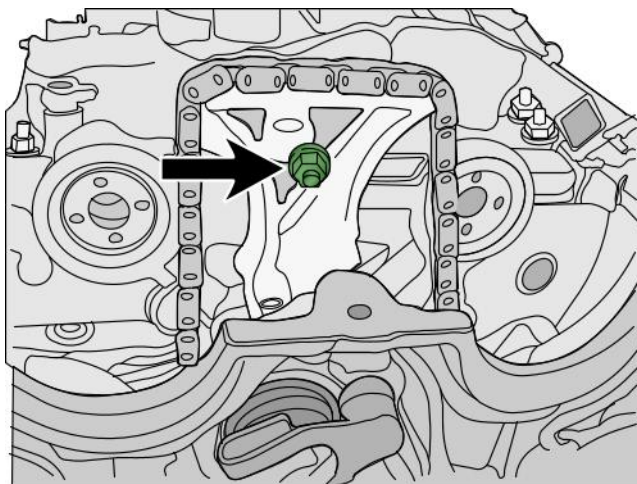
Remove the chain guide with the timing chain, guide rail and tensioner rail

Installation

Refit the chain guide with the timing chain, guide rail and tensioner rail



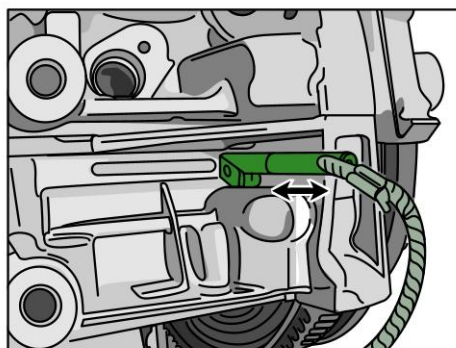
Refit the timing chain guide bolts to the cylinder head (M6 x 35: 8-9 Nm)



Refit the crankshaft hub

Use the special tools (11 5 200, 11 9 280, 11 4 360)

Ensure the crankshaft is locked (11 0 300)



Tighten the crankshaft gearwheel bolt

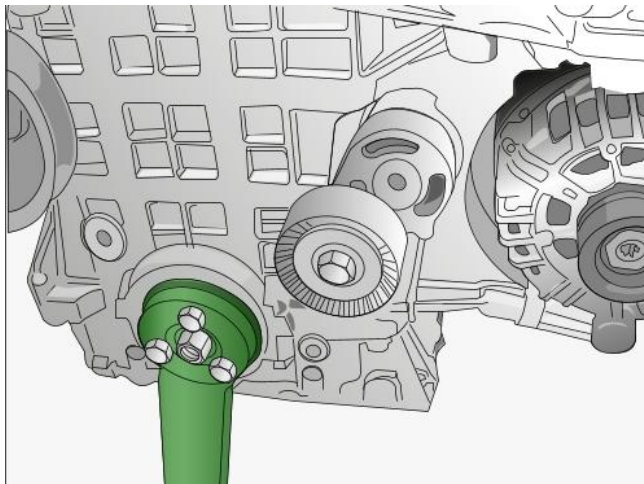
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Use new bolt(s)

Stage 1: (100 Nm)

Stage 2: (360°)

Use the special tool to counterhold (11 9 280)



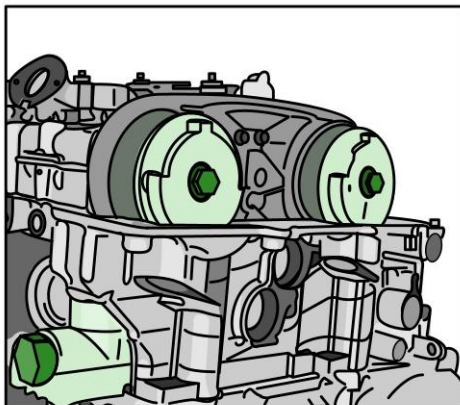
Bearing pins for guide and tensioner rails:

Refit the guide pins (M7: 14 Nm, M8: 20 Nm)

Refit the access plugs (M18 x 1.5: 25 Nm, M22 x 1.5: 50 Nm)

Refit the exhaust camshaft adjuster (AUS/EX)

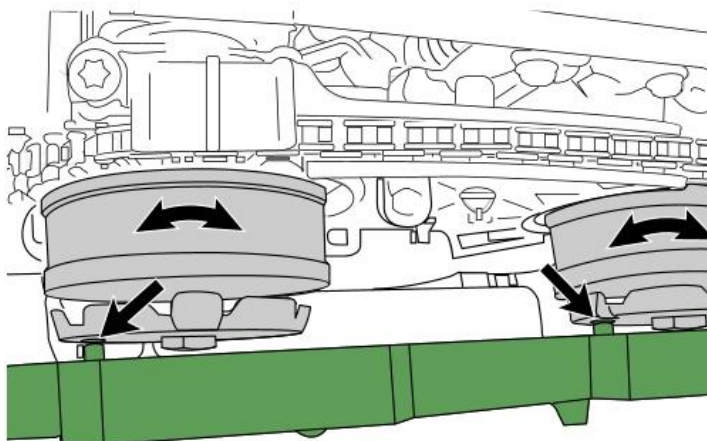
Refit the inlet camshaft timing adjuster (EIN/IN)



Tighten the camshaft gearwheel bolts lightly until there is no play

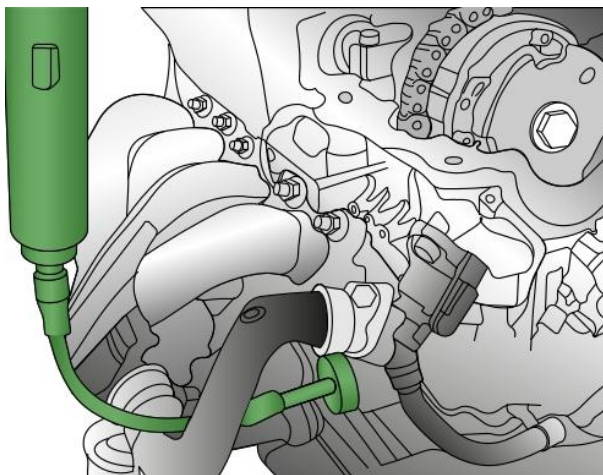
Fit the camshaft alignment tool (11 4 290)

Align the camshaft gearwheel marks as shown

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Pre-tension the timing chain (0.6 Nm)

Use the special tools (00 9 250, 11 9 340)



Tighten the camshaft gearwheel bolts

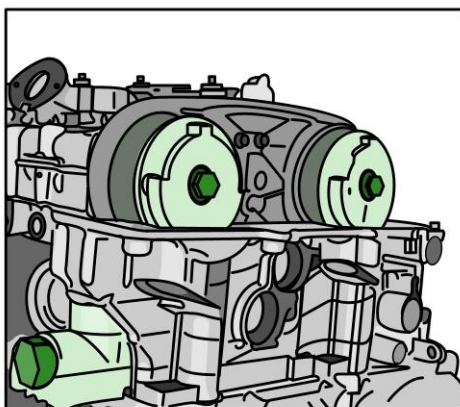
Stage 1: (20 Nm)

Stage 2: (180°)

Use new bolt(s)

Refit the chain tensioner piston (50-55 Nm)

Note: Fit a sealing ring



Remove all the locking tools

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Turn the engine 2 rotations by hand

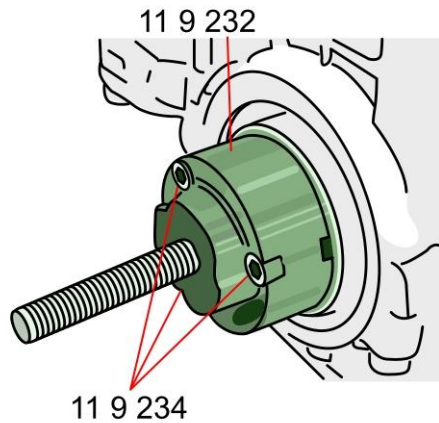
Check the timing again

Refit the ancillary drive belt tensioner

Stage 1: (25 Nm)

Stage 2: (90°)

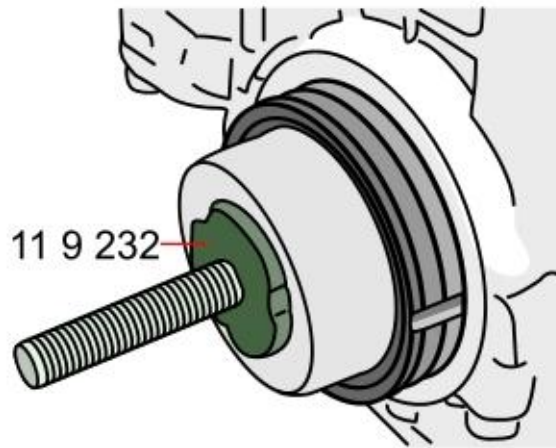
Fit the special tools (11 9 232, 11 9 234)



Apply grease

Fit the crankshaft seal over the special tool

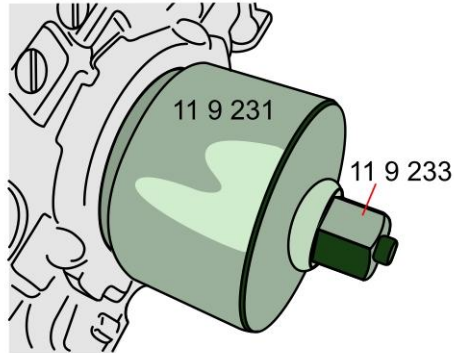
Note: The seal should be aligned with the timing cover



Refit the crankshaft front radial seal

Use the special tool (11 9 231, 11 9 233)

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Apply sealant

Fit the vibration damper (25 Nm)

Fit the ancillary drive belt

Refit the engine lower cover

Refit the spark plugs (M12 x 1.25: 23 Nm, M14 x 1.25: 30 Nm)

Refit the cylinder head cover

Use new bolt(s)

Stage 1: (7 Nm)

Stage 2: (90°)

Reconnect the cylinder head breather hose

Refit the air duct

Refit the strut brace connecting the shock absorbers

Reconnect the injection wiring

Refit and connect the ignition coils

Refit the engine upper cover

Reconnect the battery earth cable

Torque settings

Crankshaft pulley

Crankshaft pulley (Outer bolts) 35 (Nm)

Crankshaft gearwheel

Crankshaft gearwheel (Centre bolt)

Renew the bolts (Threads lightly oiled)

Stage 1 100 (Nm)

Stage 2 360 (°)

Camshaft gearwheel

Camshaft gearwheel (Renew the bolt)

Stage 1 20 (Nm)

Stage 2 180 (°)

Timing chain tensioner

Timing chain tensioner (Threads lightly oiled) 55 (Nm)

Renew the seal

Timing chain guide

Timing chain guide (Upper) 14 (Nm)

Timing chain guide (Lower) 20 (Nm)

Ancillary drive belt tensioner

Ancillary drive belt tensioner (Renew the bolt)

Stage 1 25 (Nm)

Stage 2 90 (°)

Spark plugs

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Spark plugs M12: 23 ± 3 (Nm)

Spark plugs M14: 30 ± 3 (Nm)

Valve cover

Valve cover 9 (Nm)

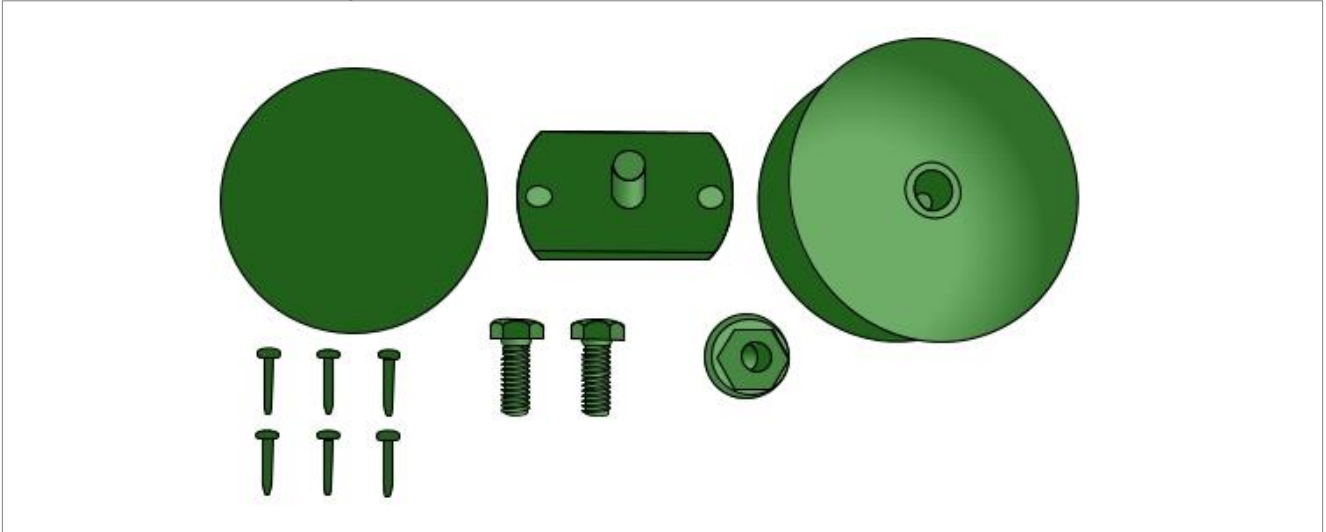
Chain guide: (M6 x 35: 8 - 9 Nm)

Access plug: (M18 x 1.5: 25 Nm)

Special tools

Extractor: (11 9 221, 11 9 222, 11 9 224, 11 0 371, 11 0 372)

Crankshaft front radial seal fitting tool: (11 9 231, 11 9 232, 11 9 233, 11 9 234)

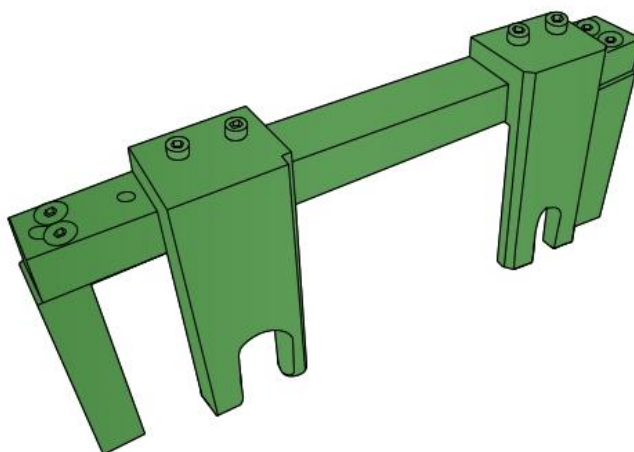


Crankshaft locking tool: (11 0 300)

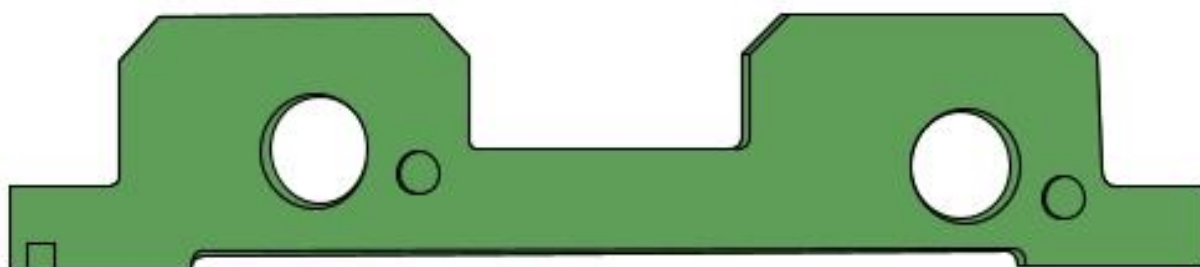


Camshaft locking tool: (11 4 281, 11 4 282, 11 4 283)

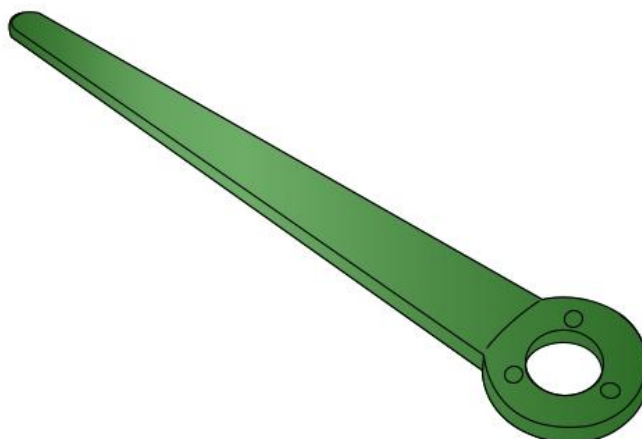
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Camshaft alignment tool: (11 4 290)

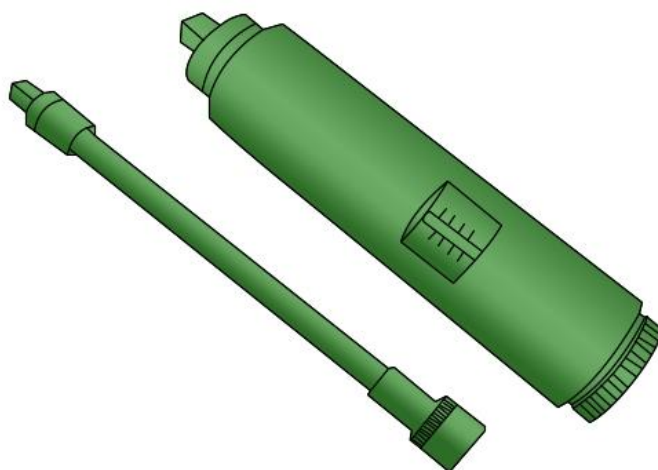


Crankshaft centering tool: (11 9 280)

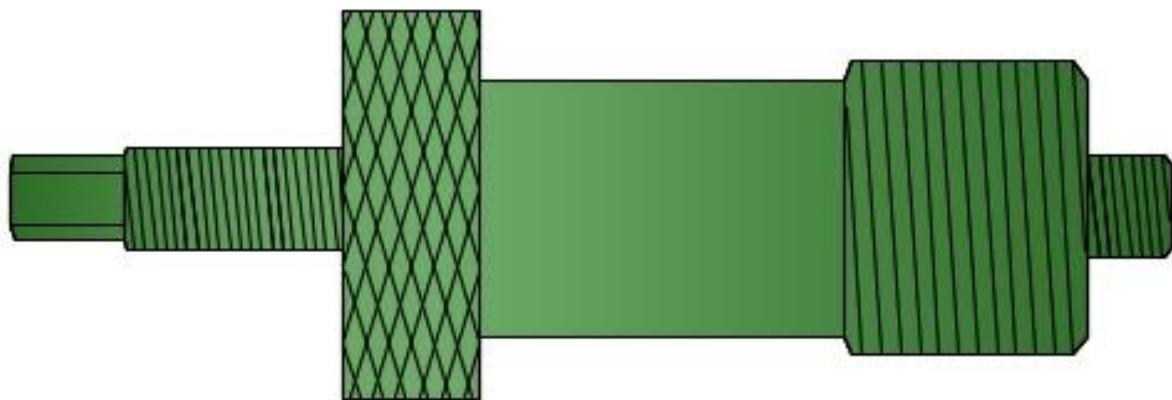


Torque wrench: (00 9 250)

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Chain tensioner: (11 9 340)



Chain tensioner locking pin: (11 3 340)

